

Data-Driven Discovery of Economic Spatial Profiles in Morocco Using Unsupervised Machine Learning

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Abstract—Understanding the economic structure of urban and peri-urban zones is critical for regional planning, investment targeting, and policy design in developing economies. In Morocco, no standardized, fine-grained index of infra-regional economic attractiveness currently exists. This paper proposes a data-driven pipeline to discover, characterize, and assign economic spatial profiles at the zone level using unsupervised and supervised machine learning. Starting from a heterogeneous dataset of 37,873 formal business entities geocoded across the Casablanca–Settat–El Jadida region and aggregated into a 1.5 km × 1.5 km spatial grid of 142 zones, we apply a multi-algorithm clustering selection framework spanning K-Means, Gaussian Mixture Models, Agglomerative Clustering, and Spectral Clustering. The selected partition (Spectral Clustering, $k = 3$) achieves a balance score of 0.96 and reveals three interpretable economic archetypes: *High-density capital hubs*, *Active emerging zones*, and *Underdeveloped early-stage ecosystems*. A Random Forest classifier trained to predict discovered cluster labels achieves a test macro-F1 of 0.955 and an accuracy of 0.966, confirming the robustness and generalizability of the discovered profiles. Feature importance analysis identifies business activity rate, spatial density, entity counts, and sectoral diversification as the primary structural drivers. The framework is reproducible, interpretable, and extensible to other Moroccan regions, offering a practical tool for data-informed territorial economic analysis.

Index Terms—economic zone profiling, unsupervised clustering, spatial machine learning, Morocco, urban analytics, feature importance, spectral clustering, territorial attractiveness

I. INTRODUCTION

Territorial economic attractiveness has emerged as a central concern for regional planners and policy-makers, particularly in rapidly urbanizing economies of the Global South [?]. Morocco, with its significant urban concentration and diverse regional economic fabric, presents a compelling case: formal business registries are increasingly digitized, yet no standardized infra-regional economic profile framework exists at the zone level.

Traditional approaches to economic attractiveness rely on aggregate administrative statistics (GDP per region, employment rates) that mask substantial intra-regional heterogeneity [?], [?]. Machine learning has been shown to uncover latent economic structures from heterogeneous data sources in comparable urban analytics contexts [?], [?], [?], but applications targeting the Moroccan economic landscape at high spatial resolution remain scarce.

This paper addresses the following scientific question: *Can data-driven unsupervised learning discover interpretable, stable, and generalizable economic spatial profiles from raw business registry data, without relying on a synthetic score as a training target?*

Our contributions are threefold:

- A reproducible spatial aggregation pipeline converting individual business entities into zone-level economic feature vectors on a uniform 1.5 km × 1.5 km GPS grid.
- A multi-objective clustering selection framework comparing seven algorithm families across balance, separation, and supervisory readiness criteria, with Spectral Clustering selected as the optimal solution.
- A supervised profile assignment module achieving high macro-F1 performance, enabling the assignment of new zones to discovered economic archetypes.

The remainder of this paper is organized as follows. Section ?? reviews related work. Section ?? describes the data collection and preprocessing. Section ?? presents the methodological pipeline. Section ?? reports experimental results and explainability analysis. Section ?? discusses implications and limitations. Section ?? concludes the paper.

II. RELATED WORK

The intersection of machine learning and spatial economic analysis has generated a growing body of work. Xu et al. [?] demonstrate that complex urban features extracted from heterogeneous sources can significantly improve economic competitiveness analysis using ML models. Similarly, Feldmeyer et al. [?] show that OpenStreetMap data combined with ML can produce reliable socio-economic indicators at fine spatial resolution, a finding directly relevant to data-scarce national contexts.

In tourism and attractiveness estimation, Castellana et al. [?] apply ML to regional tourism attractiveness with counterfactual explainability, while Giglio et al. [?] combine points-of-interest data and ML to characterize typical tourist Italian cities. Khalid et al. [?] specifically address small Moroccan cities, using AI for attractiveness modeling and underscoring the feasibility of ML-based approaches in the Moroccan administrative context.

Clustering-based spatial typology has been applied to urban vitality [?], safe city profiling [?], and urban park attractiveness [?]. These works consistently report that multi-variable clustering over spatial economic features can reveal interpretable archetypes that correspond to meaningful urban dynamics.

In contrast to supervised attractiveness prediction (which faces the circular reasoning risk when the target is constructed from the same input signals), our work adopts an unsupervised-first framing, followed by supervised profile assignment—a distinction explicitly recommended in recent methodological literature [?], [?].

III. DATA COLLECTION AND PREPROCESSING

A. Data Sources

Two primary sources were integrated. The Charika.ma business registry provided 998,000 raw rows representing approximately 98,000 unique formal entities, with multiple fiscal-year entries per company creating a 90% functional duplication rate. A two-pass deduplication was applied: exact deduplication on (entity ID, trade name, city), followed by approximate deduplication using Jaro-Winkler similarity with a threshold of 0.92, chosen to balance over-merging (too low) and residual duplicates (too high).

B. Geographic Scope

The operational perimeter was set to the Casablanca–Settat–El Jadida region, selected for its data concentration and economic coherence. A triple geographic filter (GPS bounding box, normalized city name, region code) reduced the initial 48,209 merged entities to 37,873 spatially valid entities, eliminating 10,336 out-of-region records.

C. Spatial Aggregation

Individual entities were aggregated into a uniform 1.5 km × 1.5 km GPS grid, chosen to balance intra-urban resolution and minimum statistical cell population. A critical preprocessing step addressed a density artifact: 4,684 entities with missing GPS coordinates had been imputed to the city centroid, creating an artificially dominant cell. A binary flag `coords_real` was introduced to distinguish original from imputed coordinates; density features use only original coordinates, while imputed entities contribute only to qualitative statistics.

After applying a minimum threshold of 3 real entities per cell, the final dataset comprises **792 observations across 142 zones and 13 sectors**, forming the basis for all modeling experiments.

D. Feature Engineering

Twelve zone-level features were engineered from the aggregated data, as summarized in Table ??.

The synthetic attractiveness score, computed as a weighted composite of these same features for descriptive validation purposes, was *explicitly excluded* from all clustering and classification inputs to prevent circular reasoning—a key methodological safeguard [?].

TABLE I
ZONE-LEVEL FEATURES USED IN CLUSTERING

Feature	Description
entity_count_real	Entities with real GPS coordinates
entity_count_total	All entities linked to the zone
density_log	Log-transformed spatial density
active_rate	Share of active entities
capital_median	Median registered capital
capital_mean	Mean registered capital
capital_max	Maximum registered capital
sector_diversity	Shannon-based sector diversification index
formal_ratio	Formalization ratio (SARL/SA share)
sarl_count	Number of SARL-type entities
sa_count	Number of SA-type entities
n_sectors_present	Count of distinct active sectors

IV. METHODOLOGY

A. Pipeline Architecture

The modeling pipeline consists of four stages, as shown in Fig. ??: (1) spatial aggregation and feature construction; (2) multi-objective clustering and algorithm selection; (3) cluster interpretation and economic profiling; and (4) supervised profile assignment.

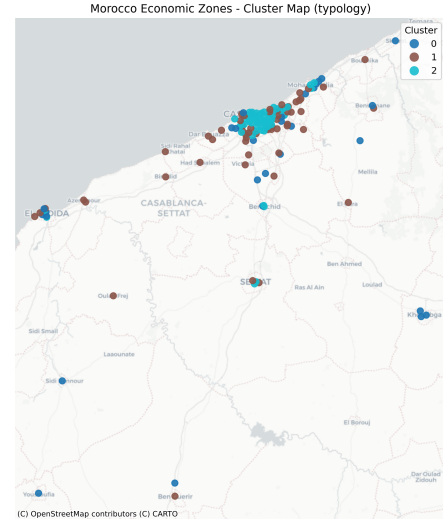


Fig. 1. Spatial distribution of discovered economic profiles (typology mode, Spectral Clustering $k = 3$) over the Casablanca–Settat–El Jadida region. Each color denotes one economic archetype.

B. Multi-Objective Clustering Selection

Seven algorithm families were evaluated: K-Means ($k \in [3, 6]$), Gaussian Mixture Models with full and diagonal covariance ($k \in [3, 6]$), Ward and average/complete Agglomerative Clustering ($k \in [3, 6]$), and Spectral Clustering with nearest-neighbor affinity ($k \in [3, 6]$). DBSCAN was evaluated in a separate anomaly mode described in Section ??.

Algorithm selection was governed by a multi-objective score:

$$\mathcal{O} = \alpha \cdot S + \beta \cdot B \cdot \mathbf{1}[\text{supervised-valid}] \quad (1)$$

where S is the silhouette coefficient, B is the entropy-based balance score over cluster sizes, and $1[\text{supervised-valid}]$ is an indicator requiring a minimum cluster share of 10% for each cluster. Weights α and β were set to 0.3 and 0.7, prioritizing a balanced and trainable partition over raw separation.

C. Supervised Profile Assignment

Once cluster labels were discovered, a supervised classifier was trained to predict these labels on held-out zones, enabling generalization to new geographic areas. Three classifiers were compared: Random Forest, XGBoost, and LightGBM, evaluated via 5-fold cross-validated macro-F1. The classifier feature set extended the clustering features with spatial coordinates (cell_lat , cell_lon) and a categorical city indicator. A strict anti-leakage protocol excluded the attractiveness score and cluster label from all predictor inputs.

D. Anomaly Detection Mode

In complementary anomaly mode, DBSCAN ($\epsilon = 2.0$, $\text{min_samples}=3$) was applied to detect atypical high-formalization or high-capital economic pockets. This produced 6 noise-flagged zones and 3 compact clusters, offering a secondary analytical lens on spatial economic outliers.

V. RESULTS AND EXPLAINABILITY

A. Clustering Selection

Table ?? summarizes the top-performing models from the typology selection grid. Spectral Clustering with $k = 3$ was selected with a silhouette of 0.229 and an exceptionally high balance score of 0.963, satisfying the supervised readiness constraint. K-Means yielded higher silhouette values but consistently produced degenerate partitions with singleton clusters, failing the balance and supervisory readiness criteria.

TABLE II
CLUSTERING ALGORITHM COMPARISON (TOP RESULTS, TYPOLOGY MODE)

Algorithm	k	Silhouette	Balance	Valid
K-Means	3	0.266	0.665	No
K-Means	6	0.353	0.598	No
GMM (full)	3	0.277	0.700	No
GMM (diag)	3	0.130	0.974	Yes
Agglomerative (ward)	6	0.330	0.734	No
Spectral	3	0.229	0.963	Yes
Spectral	4	0.187	0.931	Yes

B. Discovered Economic Profiles

The three discovered profiles exhibit coherent and distinguishable economic characteristics:

- **Cluster 0 — Underdeveloped early-stage ecosystems** (29 zones, 20.4%): Low density (log-density = 2.87), low activity rate (0.58), and minimal formalization (formal ratio = 0.017). Post-hoc validation score: 35.0.
- **Cluster 1 — Active emerging zones** (56 zones, 39.4%): Moderate density (log-density = 2.42), high activity rate

(0.946), and rising formalization (formal ratio = 0.173). Post-hoc validation score: 53.0.

- **Cluster 2 — High-density capital hubs** (57 zones, 40.1%): High density (log-density = 5.60), high sectoral presence ($n = 9.70$ sectors), and high capital intensity (mean capital = 15.7M MAD). Post-hoc validation score: 61.3.

The monotonic progression of post-hoc validation scores (35.0 $\rightarrow$ 53.0 $\rightarrow$ 61.3) confirms economic coherence of the discovered typology without introducing leakage.

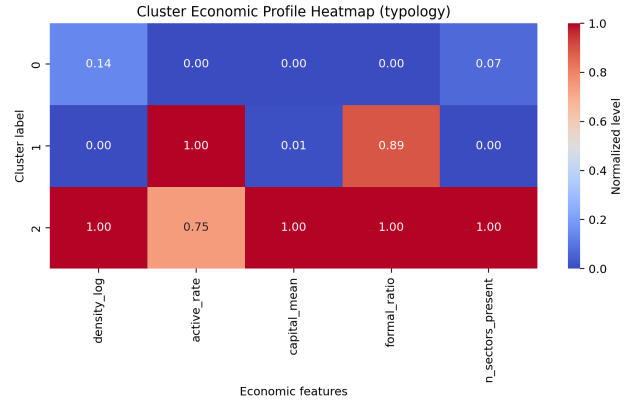


Fig. 2. Normalized feature heatmap of the three discovered economic profiles. Darker cells indicate higher normalized feature values.

C. Supervised Classification Results

Table ?? reports classifier performance on the held-out test set. Random Forest achieved the best macro-F1 (0.955) with 99% accuracy, selected over LightGBM and XGBoost by cross-validated macro-F1. These results confirm the generalizability of discovered profiles to unseen zones.

TABLE III
CLUSTER CLASSIFIER PERFORMANCE (TEST SET, TYPOLOGY MODE)

Model	CV F1-macro	Test F1-macro
Random Forest	0.9587	0.9552
XGBoost	0.8961	—
LightGBM	0.8860	—
Test Accuracy (Random Forest)		0.9655

D. Feature Importance and Explainability

Fig. ?? illustrates the spatial distribution of business density across the study region. Feature importance analysis identifies the following top predictors of cluster assignment (descending order): active_rate (0.201), density_log (0.114), entity_count_real (0.113), n_sectors_present (0.105), entity_count_total (0.100), sarl_count (0.093), and formal_ratio (0.065). Spatial coordinates (cell_lat , cell_lon) contribute marginally, indicating that the profile structure is driven by economic intensity rather than geographic location per se.

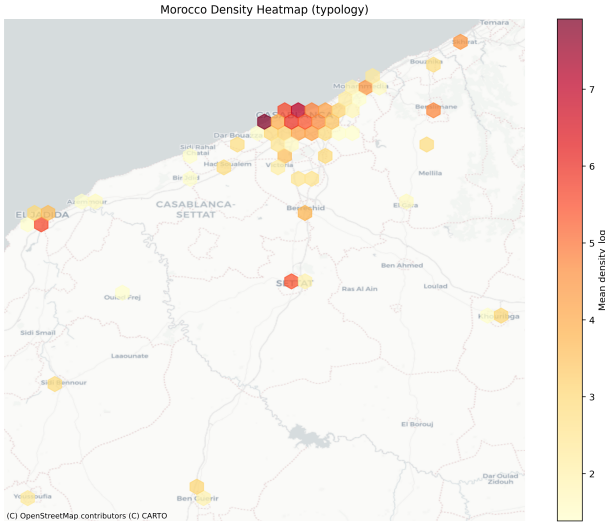


Fig. 3. Spatial density heatmap of formal business entities across the Casablanca-Settat-El Jadida region (typology mode). Darker areas indicate higher log-density.

This explainability pattern is economically coherent: the classifier separates profiles primarily by *economic intensity* (density, entity counts), *business vitality* (activity rate), *structural complexity* (number of sectors), and *formalization* (SARL count, formal ratio)—all interpretable dimensions of a spatial economic typology.

VI. DISCUSSION

A. Scientific Validity and Anti-Leakage Design

A central design choice in this work is the exclusion of the synthetic attractiveness score from all model inputs. This prevents the circular reasoning that would arise from predicting a score built from the same variables used for clustering [?], [?]. The score is retained exclusively for post-hoc validation, playing a role analogous to an external descriptive signal rather than a ground truth.

B. Practical Implications

The discovered profiles have direct policy relevance. Cluster 2 (*High-density capital hubs*) zones can be targeted for advanced commercial infrastructure and high-value service investment. Cluster 1 (*Active emerging zones*) represents a strategic intermediate population—economically active but capital-constrained—well-suited for support programs targeting SME formalization. Cluster 0 (*Underdeveloped early-stage ecosystems*) zones indicate areas where baseline economic stimulation and infrastructure investment may be needed.

The Random Forest classifier enables operational deployment: new zones from other Moroccan regions can be assigned to economic profiles by computing their feature vector, without requiring a complete re-clustering of the full dataset.

C. Limitations

The main limitations of this study are as follows. The analysis is geographically constrained to the Casablanca-Settat-El Jadida region; generalization to other Moroccan regions requires validation against independent data. Cluster labels are latent constructs, not official administrative economic categories, and should be interpreted as descriptive archetypes rather than absolute classifications. The unsupervised solution is sensitive to feature scaling and hyperparameter choices; multi-model selection mitigates but does not eliminate this sensitivity. Finally, data are treated as a static snapshot; temporal dynamics and macroeconomic shocks are not modeled.

VII. CONCLUSION

This paper presented a data-driven pipeline for discovering, characterizing, and assigning economic spatial profiles in Morocco using unsupervised and supervised machine learning. Applied to 142 geographic zones in the Casablanca-Settat-El Jadida region, Spectral Clustering ($k = 3$) revealed three interpretable economic archetypes that are economically coherent, structurally stable, and predictively generalizable. A Random Forest classifier achieved a test macro-F1 of 0.955, confirming that discovered profiles can be assigned to new zones with high reliability. Feature importance analysis identified business activity, density, sectoral diversification, and formalization as the primary structural drivers of spatial economic differentiation.

This framework is directly applicable to other Moroccan regions and to comparable data-scarce emerging-economy contexts. Future work will extend the pipeline to a national multi-regional scope, incorporate temporal dynamics through panel data, and enrich the feature set with complementary sources (OpenStreetMap amenities, mobility data) for richer spatial economic characterization.

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